

# HEALTH, SAFETY & ENVIRONMENT

Module : Environmental Management

## Historical Environmental Events

- 1945 Atom bombs at Hiroshima & Nagasaki – nuclear radiation
- 1950's Mercury poisoning, Japan
- 1967 Torrey Canyon oil tanker disaster, Scillies
- 1969 Friends of the Earth formed
- 1970 UK Royal Commission on Environmental Pollution
- 1972 UN Stockholm conference on the Environment
- 1979 Near melt-down at Three Mile Island
- 1984 Bhopal Leakage of MIC gas – over 3000 killed
- 1986 Sandoz fire pollutes River Rhine
- 1986 Chernobyl, Russia
- 1987 Brundtland Report published
- 1989 Exxon Valdez, oil tanker spill
- 1990 Shell pay £1M for Mersey pollution
- 1992 UN Earth Summit

## Historical Environmental Events

- 1992 BS 7750 draft issued
- 1995 Public outcry as Shell plan to dump the Brent Spar in the North Atlantic
- 1996 ISO 14001 issued
- 1997 Climate Treaty Kyoto (Not much success)
- 2000 Climate Convention The Hague
- 2002 World Summit on Sustainable Development, Johannesburg
- 2004 ISO 14001 revised
- 2009 Climate Change Conference, Copenhagen
- 2011 Tsunami in Fukushima, Japan
- 2015 Paris Climate Summit (ended 12-1.2015) 196 participating countries committed to limit amount of GHG to below 2° C by 2050.

Before we proceed further, shall we look at some ground reality of today ?

- ❑ As per the report of IPCC (The International Panel on Climate Change) published in 2014,
  - global warming has increased by  $0.85^{\circ}\text{C}$  in the time span of 1880-2012
  - Total  $\text{CO}_2$  accumulation in the atmosphere from 1850 to 2011, 40% added in the last four decades.
  - Direct correlation with rapid industrialisation, increased fossil fuel based economy and associated pollution
- ❑ Global warming has resulted in
  - increased acidity of oceans and the soil
  - Melting down of arctic ice at 4% per decade
  - Receding of glaciers
  - Sea level risen by 19 cm in 1901-2010
- ❑ Ozone hole since 1985 has caused skin diseases
- ❑ Environmental pollution has lead to deficiency of Vitamin D in human beings and consequent diseases.

## Some hard facts about GHG (Green House Gases) emissions:

- Steel Industry 7%
- Power Sector 37%
- Agricultural sector 17%

### This trend will lead to?

- Changing landscapes, displacement of millions
- Extinction of species
- More heat related & UV ray diseases
- Increased risk of drought and floods
- Declining crop yields
- Agricultural produce contains contaminants like heavy metals
- Increased instances of diseases like asthma due to air contamination.

**“What was once unthinkable is now unstoppable. The private sector is already increasingly in a low-emissions future. The solutions are increasingly affordable and available, and many more are poised to come especially after the success of Paris.”**

**Secretary General  
United Nations  
Dec 2015**

## Key areas of environment affected

- Air
- Water
- Land
- Noise
- Ergonomics
- Waste Management

## Protection of environment will lead to:

- Minimise harmful effects on environment caused by its activities
- Achieve continual improvement of its environmental performance
- Sustainable development compatible with diverse cultural, social, and organizational frameworks and systems of management.

**HAVE YOU EVER THOUGHT OF THESE ISSUES FOR THE PRESENT & THE FUTURE???**  
**WHAT TYPE OF EARTH ARE WE GOING TO LEAVE FOR OUR NEXT GENERATIONS!!!!!!**



# EMS – A Concept



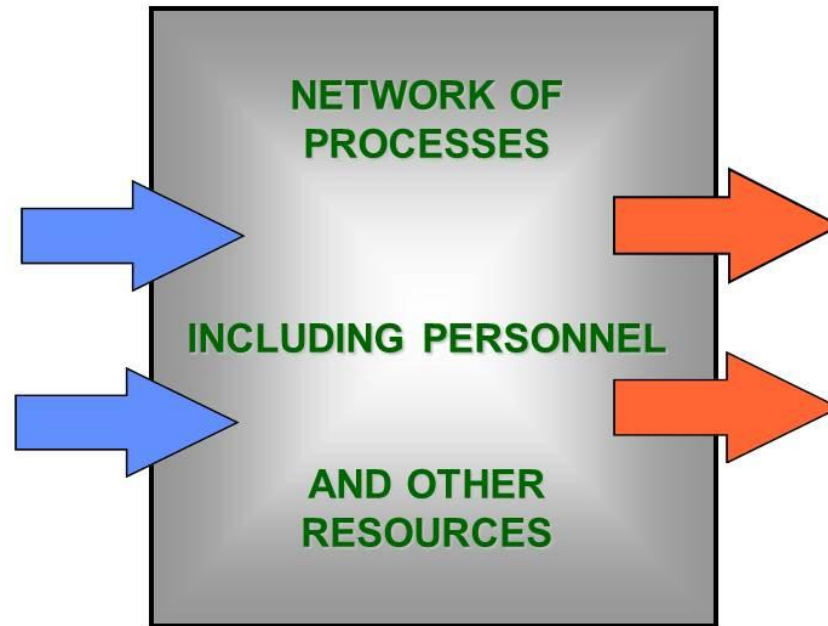
## ORGANISATION



### INPUT

**SUPPLIES**  
RAW MATERIALS  
AND  
INTERMEDIATES

**RESOURCES**  
ENERGY,  
WATER,  
LAND,  
AIR, etc,

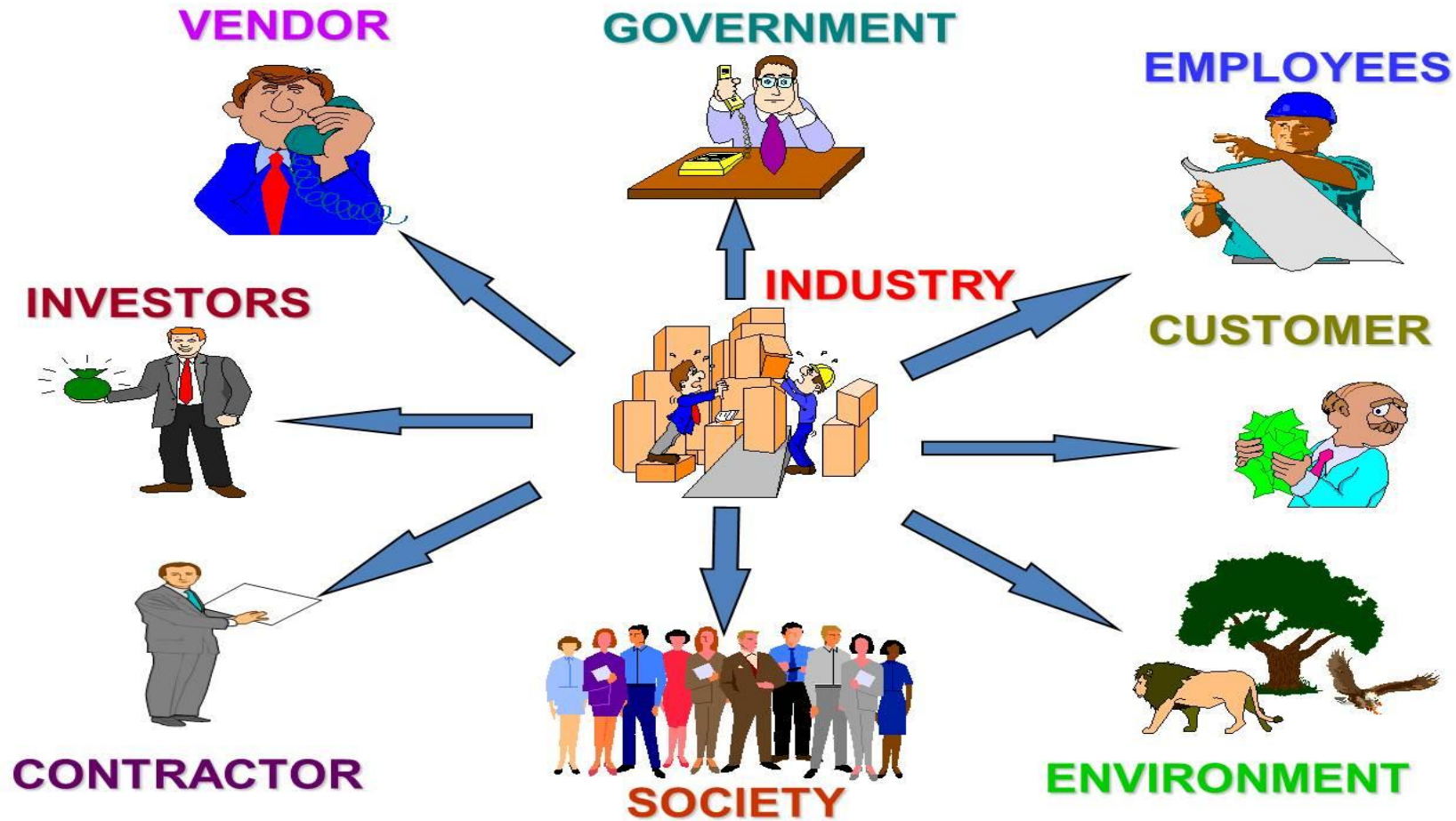


### OUTPUT

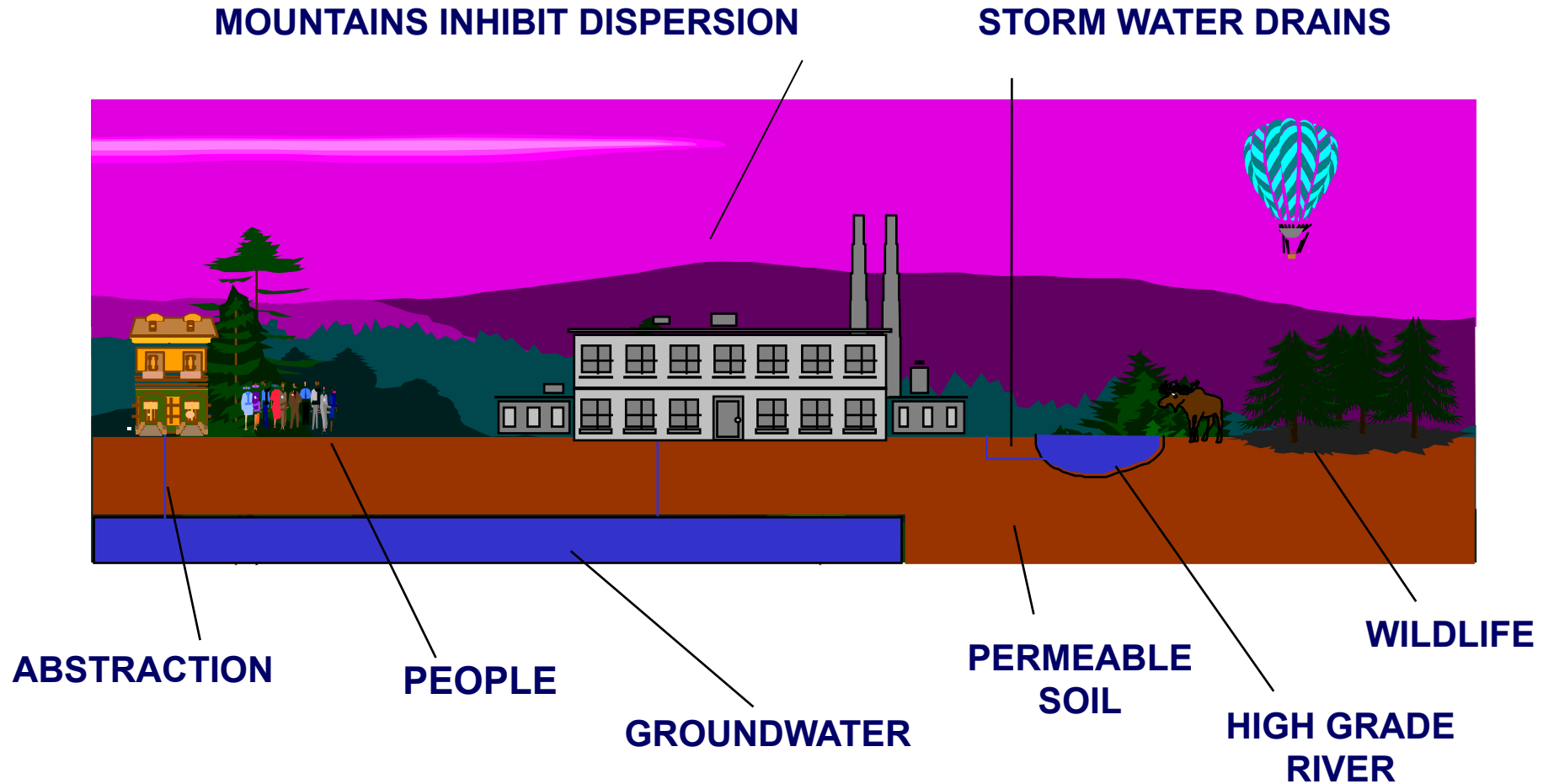
**PRODUCTS**  
CONFORMING TO  
REQUIREMENTS  
ISO 9000

**INCIDENTAL**  
PRODUCTS, WASTES,  
EMISSIONS, REJECTED  
PRODUCTS, RISKS  
AND OPPORTUNITIES  
ISO 14000

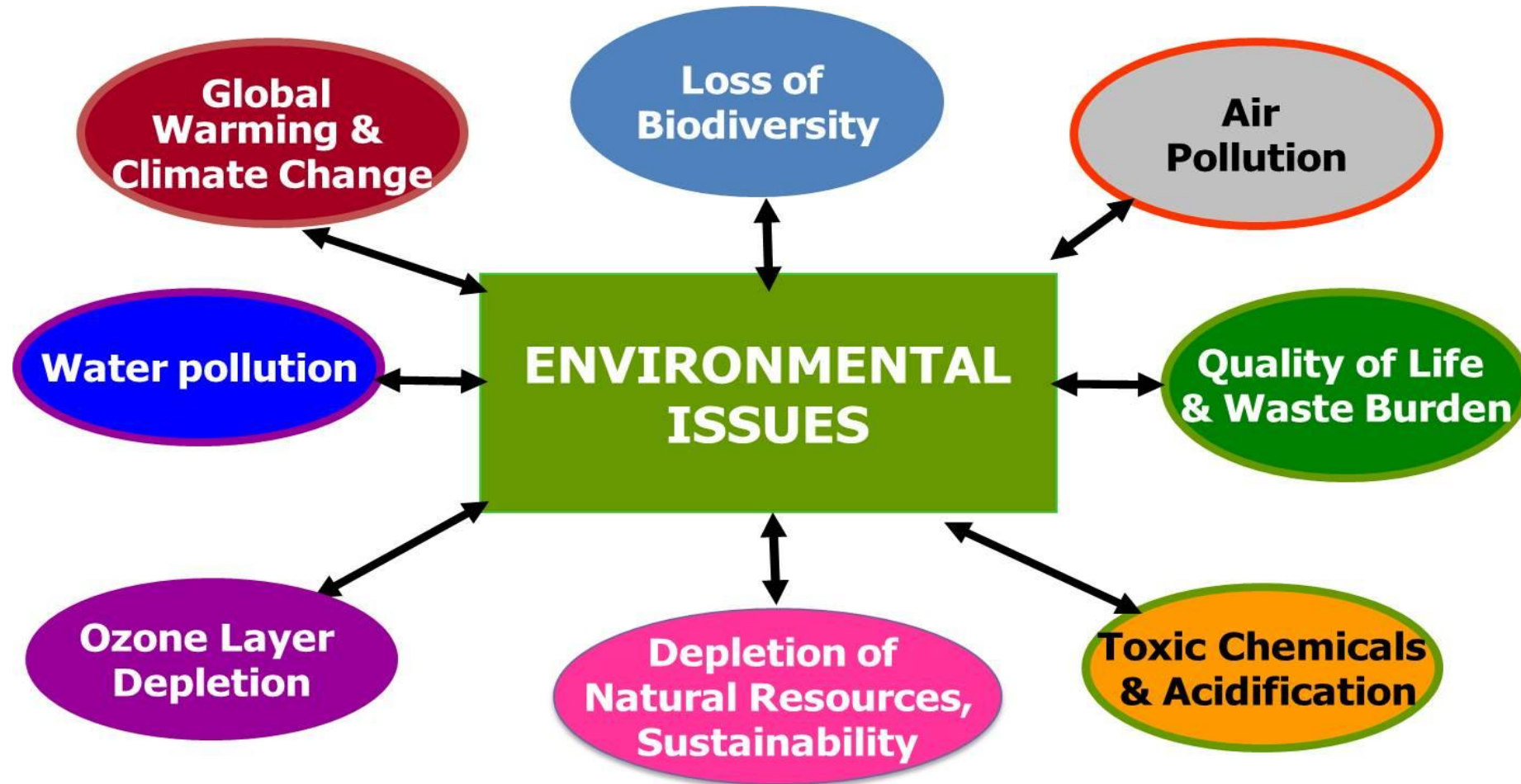
# INDUSTRY AND RELEVANT INTERESTED PARTIES



# Environmental Considerations



# Some key environmental issues of today

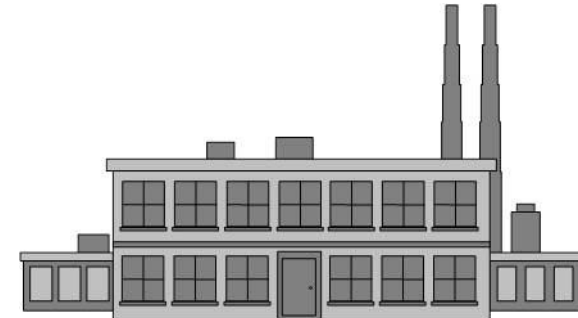


# Environmental Issues

## Air Pollution

- One fifth of the world's population breath air contaminated by pollutants in excess of WHO limits
- Acid rain, Movement of pollution, Ozone depletion, Global warming, Local air , pollution, VOC's, Vehicle emissions

**Transboundary air  
pollution convention  
Local and regional controls**



# Environmental Issues

## Water Pollution

- Effluent discharges
- Water supply
- Groundwater pollution
- Contaminated marine life
- Destruction of reefs

33% of the world's population will suffer water shortage within 30 years

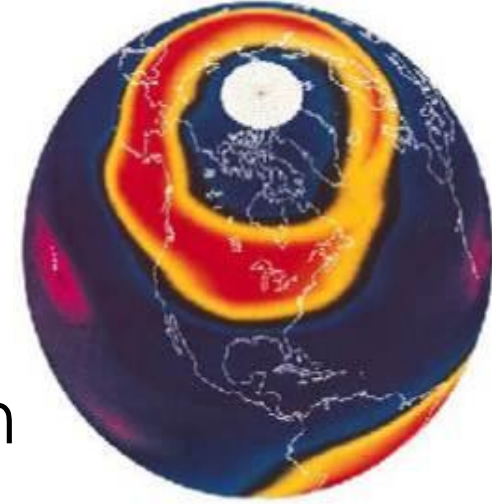


What is the Aspect and Impact?

# Environmental Issues

## Ozone Depletion

- Loss 6-8% in Northern hemisphere
- Levels critical in southern hemisph
- Skin diseases due to UV rays increase



**Montreal Protocol is an international agreement to reduce releases of ODS's, ratified by 85 countries & its follow -up**

# Global Warming

**International Convention on Climate Change in Rio, Kyoto & Stockholm Conferences, etc.**

- ❖ Global warming - Cause - Green House Gases like CO<sub>2</sub>, CH<sub>4</sub>, CFCs, N<sub>2</sub>O.
- ❖ CO<sub>2</sub> in 1810 was 280 ppm  
CO<sub>2</sub> in 2011 was 391 ppm  
CO<sub>2</sub> in 2013 crossed 400 ppm
- ❖ More than 0.45° C temperature increase in the last 85 years
- ❖ Predict ~2°C increase by 2100
- ❖ Effect - Sea level rises, variation in weather, flooding

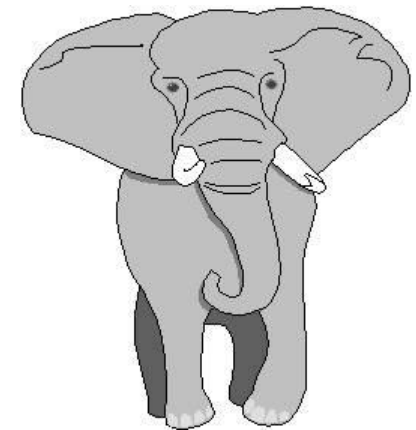


# Environmental Issues

## Loss of Biodiversity

- Diversity allows eco-systems to adapt to changes
- Tropical forests important sources of medicinal compounds
- Diversity also decreasing in developed countries

**CITES - Convention on International Trade in Endangered Species, Rio Biodiversity Convention & other conferences**



# Environmental Issues

## Depletion of Natural Resources and Sustainability

Eco-capital is unevenly distributed

- ❖ Desertification
- ❖ Population growth
- ❖ Distribution of land, water, fuels and minerals



# Environmental Issues

## Quality of Life & Waste Burden

- ❖ **NOISE**
- ❖ Nuisance (foul smell, visual)
- ❖ Visual impact
- ❖ Access to countryside
- ❖ Litter/Garbage leads to waste burden & land fills (prone to fires)
- ❖ Traffic congestion
- ❖ E - Waste



# Environmental Issues

## Toxic Chemicals

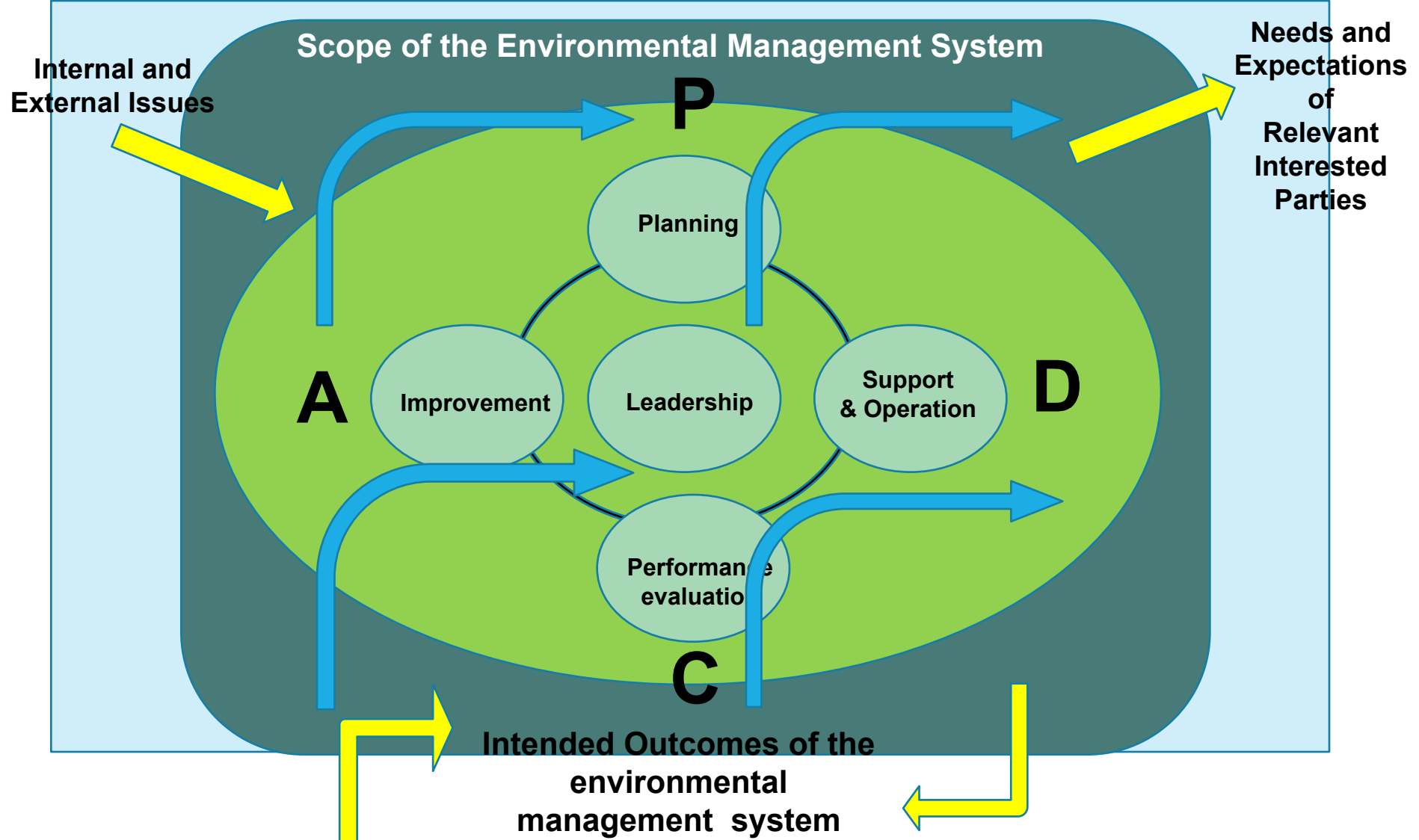
- Concentrations of DDT in Central America are 11 times higher than those of North Americans
- Toxic chemicals may bio-accumulate

Hazardous Chemicals  
Pesticides  
Insecticides  
Bio-medical Waste  
ETP Sludge



**To avoid and reduce the impact  
of these environmental issues, Is  
there a need for an effective  
Environmental Management  
System??**

# PDCA Cycle in ISO 14001:2015



# SUSTAINABLE DEVELOPMENT

**Societal expectations** for sustainable development, transparency and accountability have evolved with:

- increasingly stringent legislation,
- growing pressures on the environment from pollution,
- inefficient use of resources,
- improper waste management,
- climate change,
- degradation of ecosystems and
- loss of biodiversity.



# ENVIRONMENTAL MANAGEMENT SYSTEM

- Is intended for use by an organization seeking to manage its environmental responsibilities in a systematic manner that contributes to the environmental pillar of sustainability.
- helps an organization achieve the intended outcomes of its environmental management system, which provide value for the environment, the organization itself and interested parties.

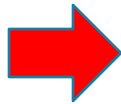


# ENVIRONMENTAL MANAGEMENT SYSTEM

## BUSINESS BENEFITS OF IMPROVING ENVIRONMENTAL PERFORMANCE, organization can:

- achieves financial and operational benefits that can result from implementing EMS that strengthen the organization's market position;
- leverage opportunities to prevent or mitigate adverse environmental impacts and enhance beneficial environmental impacts, particularly those with strategic and competitive implications.
- be used to assure interested parties that an effective environmental management system is in place.

REUSE  
REDUCE  
RECYCLE



# Air Pollution

## Causes

## Activities

|                                      |                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                      | <b>Fossil Fuel Combustion</b>                                                                                    |
| <b>Ozone</b>                         |                                                                                                                  |
| <b>Particulates</b>                  | <b>Road Traffic<br/>Construction Work<br/>Chemical Reactions</b>                                                 |
| <b>Persistent Organic Pollutants</b> | <b>Hydrocarbons Emitted by :<br/>Motor Vehicles<br/>Industrial Processes<br/>Incomplete combustion processes</b> |

# Water Pollution

## Causes

## Activities

|                    |                                                    |
|--------------------|----------------------------------------------------|
| <b>Phosphorous</b> | <b>Sewage Effluent</b>                             |
| <b>Nitrates</b>    | <b>Agriculture<br/>Geology<br/>Sewage Effluent</b> |
| <b>Oil spills</b>  | <b>Storage Activities</b>                          |

# Waste Burden

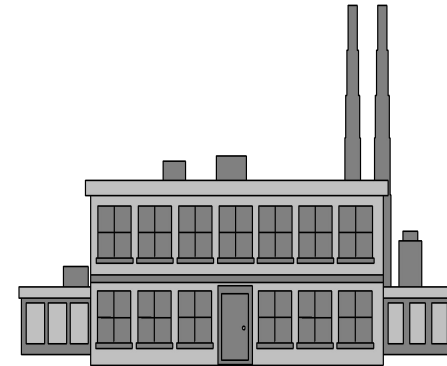
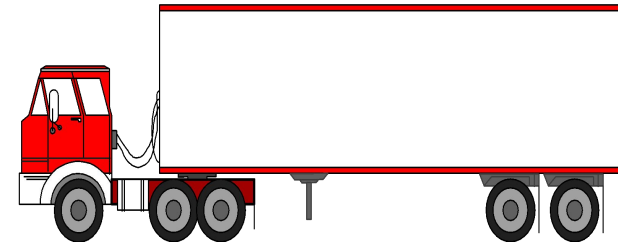
Causes

Activities

|                         |                                    |
|-------------------------|------------------------------------|
| <b>Solid and Liquid</b> | <b>Agriculture</b>                 |
|                         | <b>Mining and Quarrying</b>        |
|                         | <b>Sewage Sludge</b>               |
|                         | <b>Dredged Materials</b>           |
|                         | <b>Household</b>                   |
|                         | <b>Commercial</b>                  |
|                         | <b>Demolition and Construction</b> |
|                         | <b>Industrial</b>                  |

# Business and the Environment – Costs

- Raw Materials
- Processes
- Energy
- Waste
- Transport/Storage
- Design
- Costs of Failures  
e.g. Bhopal, Fukushima Nuclear



# Business and the Environment – Drivers

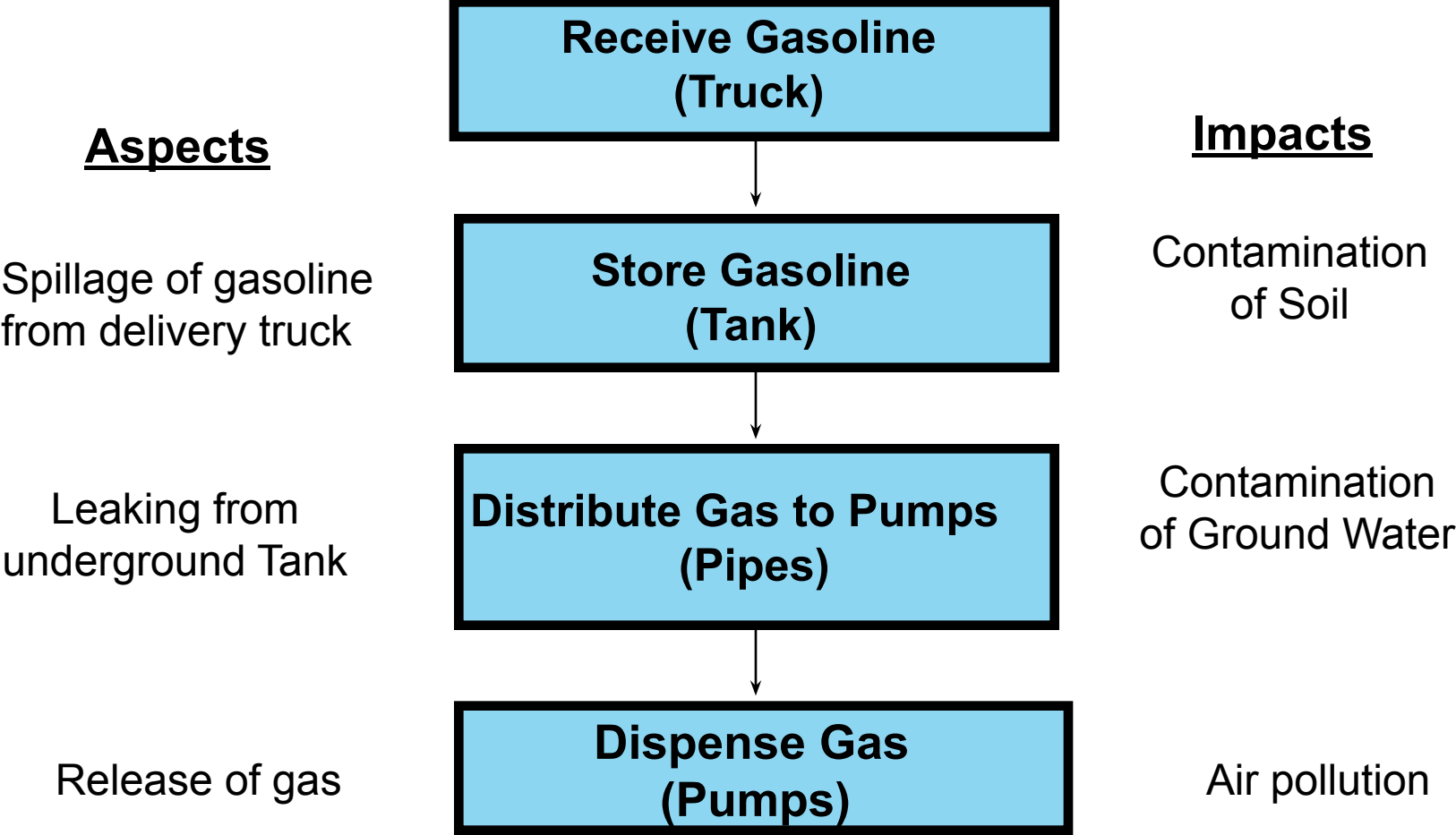
- Achievement of business objectives
- Corporate governance
- Corporate social responsibility
- Sustainability and biodiversity
- Increasing legislation
- Increased expectations and requirements of interested parties
- Increasing cost burden
- Loss of competitiveness

# Identifying and Evaluating Environmental Impacts

## Environmental impacts can be categorized

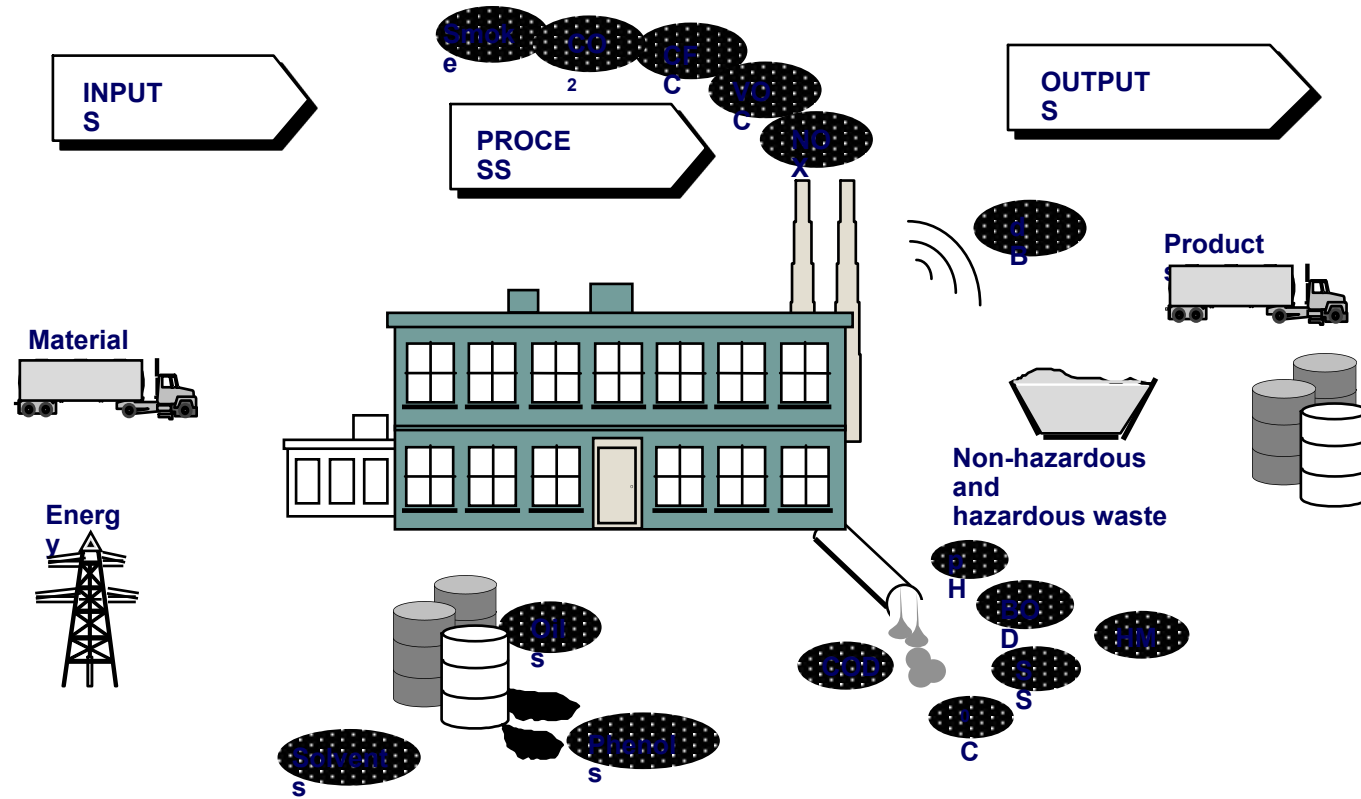
- Greenhouse Effect
- Stratospheric Ozone Depletion
- Natural Resources
- Acid Rain
- Surface Water Quality
- Bio-accumulation of Toxins
- Local Air Quality
- Waste burden
- External noise
- Soil & groundwater contamination
- Ecological habitat
- Visual intrusion

# Gas Station Example



# Inputs, Process and Outputs

Environmental impacts arise from the consumption of resources and emissions to air, water and land



# Environmental Aspects and Impacts – Other Impacts

| ASPECT             | IMPACTS                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Land contamination | Harm to animals and plants; water pollution through run-off, degradation of soil quality; groundwater pollution.                                        |
| Use of pesticides  | Harm to animals and plants; water pollution through run-off; degradation soil quality; groundwater pollution; resource depletion; ecological imbalance. |
| Site development   | Visual disturbance; noise; vehicle pollution; energy use and subsequent impacts; river pollution; groundwater pollution; resource depletion.            |
| Supply chains      | All impacts to air; land and water; resource depletion.                                                                                                 |